

Chapter 5

Pairs of Continuous R.V.

Let X, Y - two continuous r.v.

Def 1: The joint CDF of X and Y is:

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$

Remark!

The properties of joint CDF for continuous r.v. are the same as for discrete r.v.:

1. $\lim_{\substack{x \rightarrow \infty \\ y \rightarrow \infty}} F_{X,Y}(x,y) = 1$

2. $\lim_{x \rightarrow \infty} F_{X,Y}(x,y) = \lim_{y \rightarrow \infty} F_{X,Y}(x,y) = 0$

3. $F \nearrow$ (funcție crescătoare)

4. $F_{X,Y}(x,y) \in [0,1]$

5. $F_X(x) = \lim_{y \rightarrow \infty} F_{X,Y}(x,y)$

$$F_Y(y) = \lim_{x \rightarrow \infty} F_{X,Y}(x,y)$$

Def 2: The joint PDF of X and Y is a function, such as:

$$F_{X,Y}(x,y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u,v) du dv$$

Remark!

The joint PDF of X and Y is given by:

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x \partial y}$$

Properties:

1. $f_{X,Y}(x,y) \geq 0, \forall (x,y)$

2. $\iint_{\mathbb{R}^2} f_{X,Y}(x,y) dx dy = 1$
 $= \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$

Prop: if $A \subset (x,y)$, A -an event, then:

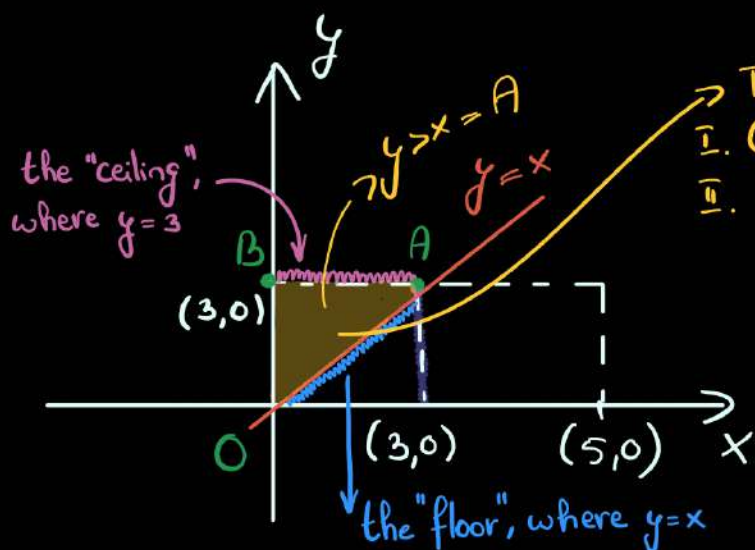
$$P(A) = \iint_A f_{X,Y}(x,y) dx dy$$

Ex: Consider that X, Y are two continuous r.v., having the joint PDF:

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{15}, & x \in [0;5], y \in [0;3] \\ \text{zero,} & \text{otherwise} \end{cases}$$

Determine $\underline{P(Y > X)}$
= A

Denote by A - the event $\Rightarrow P(A) = \iint_A f_{X,Y}(x,y) dx dy$



Two options to calculate:

- I. Calculate the area of the triangle
- II. Calculate the double integral

I. $\underline{P(A)} = \frac{1}{15} \cdot \mathcal{A}_{\Delta ABO} = \frac{1}{15} \cdot \frac{3 \cdot 3}{2} = \boxed{\frac{3}{10}}$

II. Transform the double integral into an iterative integral by calculating the boundaries of x and y :

• proj_{ox} $A = [0;3] \Rightarrow x \in [0;3]$

• But how do we calculate for y ?

y is moving between the floor and the ceiling $\Rightarrow$

$\Rightarrow y \in [OA; AB] \Rightarrow y \in [x; 3]$
 $\downarrow \quad \downarrow$
 $y=x \quad y=3$

$\Rightarrow \iint_A \frac{1}{15} dx dy = \boxed{\int_0^3 dx \int_x^3 \frac{1}{15} dy}$ - the iterative form

Def: The marginal PDFs of X and Y are:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$$

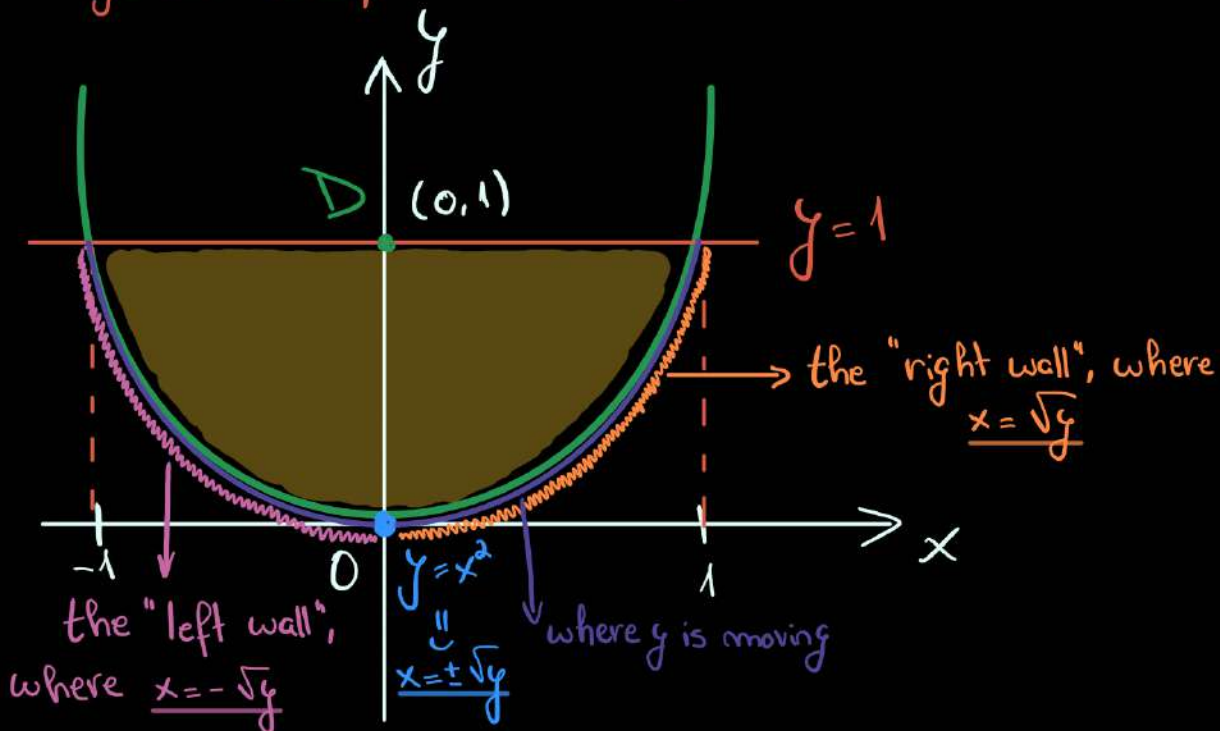
$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx$$

Ex: Consider that X, Y are two continuous r.v., having the joint PDF:

$$f_{X,Y}(x,y) = \begin{cases} \frac{5y}{4}, & x \in [-1; 1], x^2 \leq y \leq 1 \\ \text{zero, otherwise} \end{cases} \quad \xrightarrow{y=x^2}$$

Determine: $f_X(x), f_Y(y)$

Always draw the picture! it HeLPS!



We have constant domains only when we work with squares or rectangles. in the other cases, only the first integral has a constant domain.

$$\underline{f_X(x)} = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy = \int_{x^2}^1 \frac{5y}{4} dy = \frac{5}{4} \frac{y^2}{2} \Big|_{x^2}^1 = \boxed{\frac{5}{8}(1-x^4)}$$

$$\Rightarrow f_X(x) = \begin{cases} \frac{5}{8}(1-x^4), & x \in [-1;1] \\ \text{Zero, otherwise} \end{cases} \begin{array}{l} \xrightarrow{\text{proj}_{OX} D} \text{(the projection of } OX \text{ on the domain)} \\ \xrightarrow{\text{no } y \text{ here!!}} \end{array}$$

$$\underline{f_Y(y)} = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{5y}{4} dx = \frac{5y}{4} x \Big|_{-\sqrt{y}}^{\sqrt{y}} = \frac{5y}{4} (\sqrt{y} + \sqrt{y}) = \boxed{\frac{5y\sqrt{y}}{2}}$$

$$\Rightarrow f_Y(y) = \begin{cases} \frac{5y\sqrt{y}}{2}, & y \in [0;1] \\ \text{Zero, otherwise} \end{cases} \begin{array}{l} \xrightarrow{\text{proj}_{OY} D} \text{(the projection of } OY \text{ on the domain)} \\ \xrightarrow{\text{no } x \text{ here!!}} \end{array}$$

Functions of two continuous r.v.

Consider that: X, Y - continuous r.v.

$$\begin{array}{l} g(x,y) - \text{continuous function} \\ W = g(X,Y) \end{array}$$

$\Rightarrow W$ - continuous r.v.

Prop: The CDF of W is given by:

$$F_W(w) = P(W \leq w) = \iint_{g(x,y) \leq w} f_{X,Y}(x,y) dx dy$$

Prop: if $W = \max(X, Y)$, then:

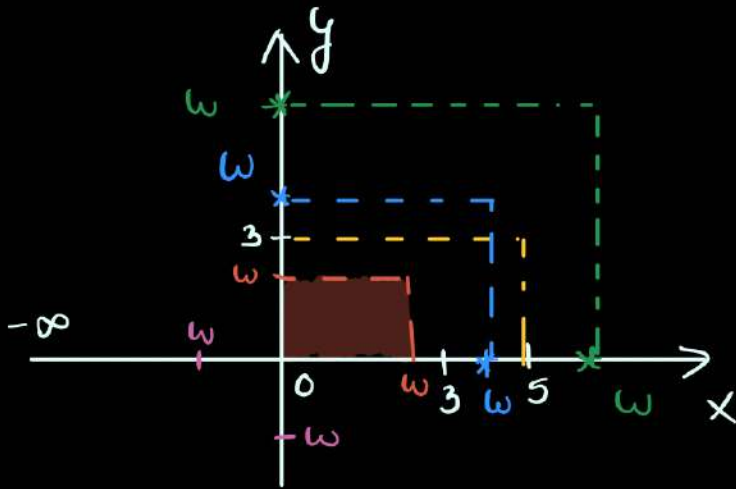
$$F_W(w) = \int_{-\infty}^w \int_{-\infty}^w f_{X,Y}(x,y) dx dy$$

Ex: Consider that X, Y are two continuous r.v., having the joint PDF:

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{15}, & x \in [0;5], y \in [0;3] \\ \text{zero,} & \text{otherwise} \end{cases}$$

Determine PDF for $W = \max(X, Y)$

$$F_W(w) = \int_{-\infty}^w \int_{-\infty}^w f_{X,Y}(x,y) dx dy$$



• $w < 0 \Rightarrow$
 $\Rightarrow F_W(w) = \int_{-\infty}^w \int_{-\infty}^w 0 dx dy = \underline{0}$

• $w \in [0;3] \Rightarrow$
 $\Rightarrow F_W(w) = \int_{-\infty}^w \int_{-\infty}^w \frac{1}{15} dy$

$$F_W(w) = \frac{1}{15} \times \frac{w}{0} \cdot \frac{y}{0} = \boxed{\frac{w^2}{15}}$$

• $w \in [3;5] \Rightarrow F_W(w) = \int_0^w dx \int_0^3 \frac{1}{15} dy = \frac{1}{15} \times \frac{w}{0} \cdot \frac{y}{0} = \frac{3w}{15} = \boxed{\frac{w}{5}}$

• $w \geq 5 \Rightarrow F_W(w) = \int_0^5 dx \int_0^3 \frac{1}{15} dy = \frac{1}{15} \times \frac{5}{0} \cdot \frac{y}{0} = 1$

$$\Rightarrow F_W(w) = \begin{cases} 0, & w < 0 \\ \frac{w^2}{15}, & w \in [0;3] \\ \frac{w}{5}, & w \in [3;5] \\ 1, & w \geq 5 \end{cases}$$

- CDF of W

$$f_W(w) = F'_W(w) = \begin{cases} \frac{2w}{15}, & w \in [0; 3) \\ \frac{1}{5}, & w \in [3; 5) \\ \text{zero,} & \text{otherwise} \end{cases} \quad \text{— PDF of } W$$

Def: For X, Y — two continuous r.v., the expected value of $W = g(X, Y)$ is:

$$E(W) = \iint_{\mathbb{R}^2} g(x, y) \cdot f_{X, Y}(x, y) \, dx \, dy$$

Remark!

The definitions and properties for $\text{cov}(X, Y)$, $r_{X, Y}$, $f_{X, Y}$ are the same as for discrete r.v.:

Def: The covariance of X and Y is:

$$\text{cov}(X, Y) = E[(X - E(X)) \cdot (Y - E(Y))]$$

The correlation of X and Y is:

$$r_{X, Y} = E(X \cdot Y)$$

Properties:

1. $\text{cov}(X, Y) = r_{X, Y} \cdot E(X) \cdot E(Y)$
2. $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \cdot \text{cov}(X, Y)$

Def: X and Y are uncorrelated $\Leftrightarrow \underline{\text{cov}(X, Y) = 0}$

X and Y are orthogonal $\Leftrightarrow \underline{r_{X,Y} = 0}$

Def: The correlated coefficient of X and Y is:

$$\rho_{X,Y} = \frac{\text{cov}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}}$$

Prop: if $Y = aX + b$, then: $\rho_{X,Y} = \begin{cases} 1, & a > 0 \\ -1, & a < 0 \\ 0, & a = 0 \end{cases}$

Conditioning by an event

Consider that: X, Y - two continuous r.v. $\rightarrow f_{X,Y}(x,y)$ - joint PDF
 $A \in (\mathcal{X} \times \mathcal{Y}), P(A) > 0$ - event

Def: The conditional joint PDF of X and Y given A is:

$$f_{X,Y|A}(x,y) = \begin{cases} \frac{f_{X,Y}(x,y)}{P(A)}, & (x,y) \in A \\ \text{zero,} & \text{otherwise} \end{cases}$$

The conditional expected value of $W = g(X, Y)$ given A is:

$$E(W|A) = \iint_{\mathbb{R}^2} g(x,y) \cdot f_{X,Y|A}(x,y) \, dx \, dy$$

The conditional variance of $W=g(X,Y)$ given A is:

$$\text{Var}(W|A) = E(W^2|A) - [E(W|A)]^2$$

Ex: Consider that X, Y are two continuous r.v., having the joint PDF:

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{15}, & x \in [0;5], y \in [0;3] \\ \text{zero,} & \text{otherwise} \end{cases}$$

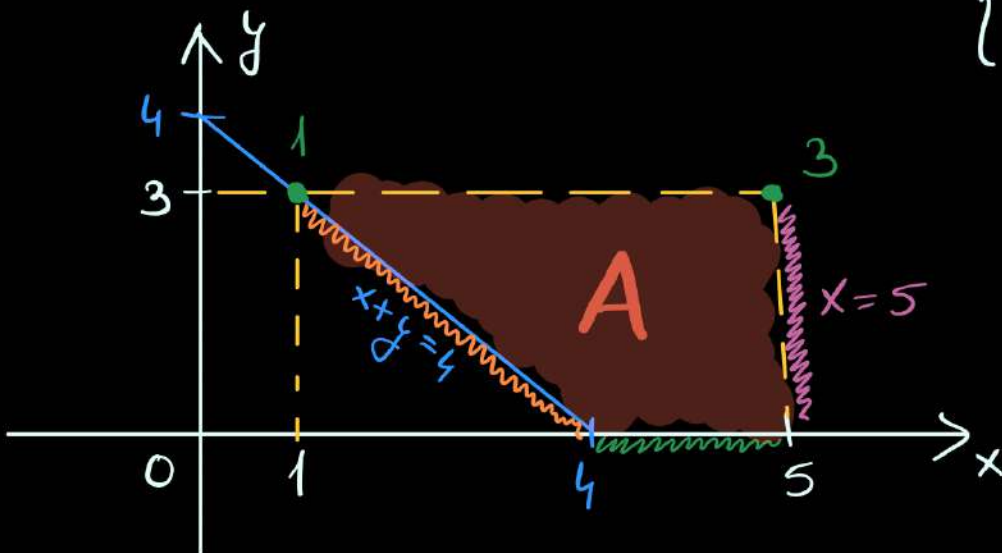
Determine: a. $f_{X,Y|A}(x,y) = ?$ $A = \{X+Y \geq 4\}$

b. $E(W|A) = ?$ $W = X \cdot Y \Rightarrow g(x,y) = x \cdot y$

$$\text{a. } P(A) = \iint_A f_{X,Y}(x,y) dx dy$$

$$A: X+Y \leq 4 \begin{cases} (4;0) \\ (0;4) \end{cases}$$

$$\begin{cases} x+y=4 \\ y=3 \end{cases} \Rightarrow (1;3)$$



Look after both projections on OX and OY and choose the one that is easier:

- $\text{proj}_{OX} A = [1; 5] \Rightarrow \underline{x \in [1; 5]}$

$$y \begin{cases} x \in [4; 5] \Rightarrow \underline{y \in [0; 3]} \\ x \in [1; 4] \Rightarrow \underline{y \in [4-x; 3]} \end{cases}$$

- $\text{proj}_{OY} A = [0; 3] \Rightarrow \underline{y \in [0; 3]}$

$$x - y \in [1; 4] \Rightarrow \underline{x \in [4-y; 5]}$$

this case is easier
!!
we will use this one

$$\Rightarrow P(A) = \int_0^3 dy \int_{4-y}^5 \frac{1}{15} dx = \frac{1}{15} \int_0^3 (x / 4-y)^5 dy = \frac{1}{15} \int_0^3 \underbrace{(5-4+y)}_{=1+y} dy$$

$$\underline{P(A)} = \frac{1}{15} \left(y + \frac{y^2}{2} \Big|_0^3 \right) = \frac{1}{15} \left(3 + \frac{1}{2} \right) = \boxed{\frac{1}{2}}$$

$$\Rightarrow f_{X,Y|A}(x,y) = \begin{cases} \frac{1}{15} \\ \frac{1}{2} = \frac{2}{15}, & (x,y) \in A \\ \text{zero, otherwise} \end{cases}$$

b. $\underline{E(W|A)} = \iint_{\mathbb{R}^2} g(x,y) \cdot f_{X,Y|A}(x,y) dx dy = \iint_{\mathbb{R}^2} x \cdot y \cdot f_{X,Y|A}(x,y) dx dy$

$$= \iint_{\mathbb{R}^2} x \cdot y \cdot \frac{2}{15} dx dy = \frac{2}{15} \int_0^3 dy \int_{4-y}^5 x \cdot y dx = \frac{2}{15} \int_0^3 \left(\frac{x^2}{2} \cdot y \right) \Big|_{4-y}^5 dy$$

the integral
is over the
same area

$$= \frac{1}{15} \int_0^3 y [25 - (4-y)^2] dy = \dots = \boxed{\frac{123}{20}}$$